

Decoding Emotions: Learning Feelings from Textual Data

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Abstract

Automatic emotion recognition from text is a fundamental challenge in natural language processing, with applications ranging from mental health monitoring to human-computer interaction. In this report, we propose and evaluate transformer-based models for emotion classification from text, using different classification head architectures. In particular, we propose a pretrained RoBERTa model to extract contextualized representations of text, and apply different pooling strategies alongside deep classification layers to improve emotion prediction performance. Our experimental results show that the proposed architectures achieve superior performance compared to the baseline model, with improvements of up to 20.94% in micro-F1 and 27.89% in macro-F1 score, even outperforming human capabilities up to 26% on average. Our results emphasize the effectiveness of transformer-based architectures in capturing nuanced emotional expressions and highlight the importance of choosing a fitting classification approach for detecting emotions from textual data.

1 Introduction

Emotions play a central role in the daily life of every human being. They influence our decision-making (George and Dane, 2016), interpersonal relationships (Fitness, 2015) and our general well-being (Fredrickson and Joiner, 2002). Humans are excellent at identifying emotional states whenever we have access to sensory readings like facial expressions, voice level and used vocabulary (Ferretti and Papaleo, 2019). However, if we only have access to language, identifying emotion becomes more difficult. Although it has been shown that it is possible to express emotions through lexical elements and grammatical structures (Schwarz-Friesel, 2015), interpretation remains difficult - for both humans and machines.

Nowadays, the transformer architecture (Vaswani, 2017) enables human-level language processing by leveraging self-attention to capture nuanced emotional expressions with wide-ranging dependencies and contextual flexibility. This capability is critical for robust emotion recognition models that have diverse applications, particularly in digital health monitoring. As social networks serve as the primary platform for emotion expression, leveraging their rich textual data together with advanced emotion recognition can help in early detection of mental health risks and potentially prevent stress, anxiety and depression (Park et al., 2013).

In this work, we study the capabilities of transformer-based models to accurately predict emotions and their intensities from emotion-annotated textual data. Therefore, we make the following contributions:

1. We propose five model architectures for emotion detection and intensity classification in Section 3. Further, we describe the applied training and preprocessing strategy in the same chapter and evaluate the results of our approach in Section 4.
2. In Section 4.3, we describe and evaluate a human-machine interface to let our model compete against a human for text based emotion classification.
3. We analyze the explainability of our best performing models using SHAP (Shapley, 1953) in Section 4.4.

An overview of our experimental pipeline is depicted in Figure 1.

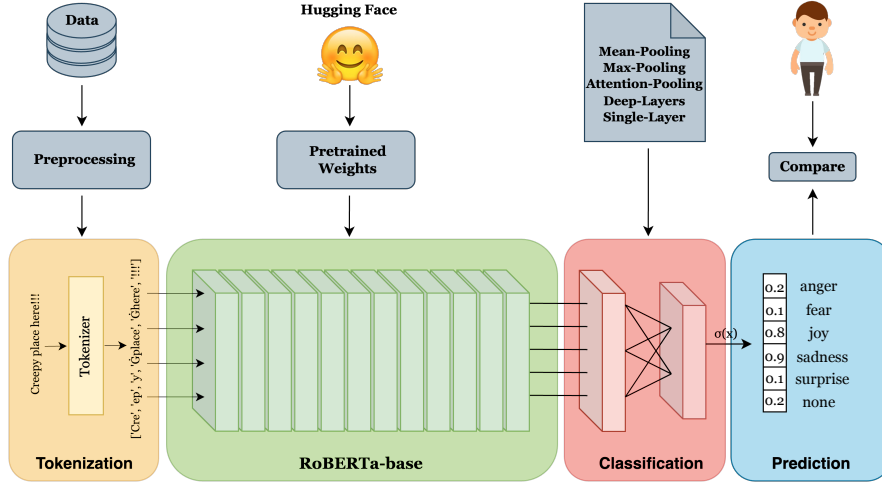


Figure 1 Overview of the emotion detection pipeline. The dataset from the SemEval competition is preprocessed and tokenized. The tokenized data is processed by a pretrained RoBERTa model. The contextualized token embeddings of the last RoBERTa layer are then used for emotion classification using different classification approaches. Finally, the model’s multi-emotion prediction capability is compared to human performance.

2 Related Work

Models of Emotion. Models of human emotion can be divided in two categories: dimensional and classification models (Imran et al., 2020). Ekman (Ekman, 1992) proposed a classification model consisting of the six basic emotions *anger*, *disgust*, *fear*, *happiness*, *sadness*, and *surprise*. Another well known classification model by Plutchik (Plutchik, 1980) features eight distinct emotions. Dimensional models characterize emotions by defining where they are located in two- or three-dimensional spaces (Lee et al., 2012). Examples for dimensional models are the Positive Affect Negative Affect Scale (PANAS) (Watson et al., 1988) and the Pleasure, Arousal and Dominance (PAD) model (Mehrabian, 1996). In this work, we build upon Ekman’s categorial model (Ekman, 1992), due to the categorized labeling of our dataset. Compared to Ekman’s six emotions, we exclude *disgust* due to a lack of samples in our data and have introduced a new category called *none*, which characterizes the absence of any emotion. Further, we use the three stages *light*, *medium* and *strong* to measure the intensity of each emotion except *none*.

Text-to-Emotion Models. Early work in the field of text-based emotion detection proposed solutions relying on keyword- or lexical-based approaches (Shivhare et al., 2015; Wang and

Pal, 2015; Bandhakavi et al., 2017). Later on, researcher started to use conventional machine learning algorithms for classifying text. Dominant among those approaches are Support-Vector-Machines (Hajar et al., 2016; Rana and Kolhe, 2015) and Naive-Bayes Classifier (Junianto and Rachman, 2019). Due to their limitations in textual understanding, the landscape shifted towards learning based emotion classification. Multiple studies have focused on using the transformer architecture to classify emotions based on text. Yang et al. (Yang et al., 2019) trained a BERT (Devlin, 2018) based model to classify emotions from textual dialog data into four distinct categories. They retrieved contextualized dialogue embeddings from the BERT model, which were then passed to a max-pooling layer to retrieve multiple single utterance embeddings that are fed into feed-forward classification heads for multiple predictions. In comparison, we only use a single sentence as input for our model by having more emotion categories in a multi-label setup. Multiple related work proposed to use both transformers and recurrent networks for emotion classification (Jiao et al., 2019; Huang et al., 2019; Vlad et al., 2019; Wang and Mine, 2023). Huang et al. (Huang et al., 2019) proposed a work to classify textual conversations into four emotion categories. In comparison to the previous approach, they used a specialized utterance-encoder and context

Long short-term memories (LSTMs) (Hochreiter, 1997) to extract features that are passed through a multi-head self-attention layer before classification. In comparison to Yang et al. (Yang et al., 2019) they get a single classification result from the proposed model. We obtain a single result per inference step using a single transformer model with a custom classification head, eliminating the need for specialized sentence encoders or additional models. Mathew et al. (Mathew and Prakash, 2023) uses RoBERTa and weighted pooling to classify text into multiple distinct emotion categories. We use embedded tokens from the last hidden layer of RoBERTa for pooling and classification, while they aggregate the outputs from the last four layers. Finally, Mathew et al. used a single pooling strategy, while we evaluate multiple pooling strategies alongside classification approaches without pooling. For an overview of general developments and applications of text-to-emotion models, we refer to (Acheampong et al., 2021; Al Maruf et al., 2024; Chutia and Baruah, 2024).

Explainability Analysis. Explainable AI (XAI) aims to enhance the transparency, interpretability, and comprehensibility of AI models, ensuring that their decisions are trustworthy, verifiable, and improvable. Explanations help identify model weaknesses and biases, with various methods available to explain black-box models, such as SHAP (Lundberg and Lee, 2017). SHAP is a game-theoretic approach that employs Shapley values (Shapley, 1953) to provide a consistent measure of feature importance by attributing output contributions to input features. As a model-agnostic method, SHAP does not require prior knowledge of the model itself. However, comparing model explainability remains challenging due to the absence of standardized, objective metrics (Fantozzi and Naldi, 2024). Additionally, Mitra et al. (Mitra and Gilpin, 2023) found no correlation between a model’s explainability and its accuracy. Similarly, Roy et al. (Roy et al., 2022) observed that methods such as SHAP and LIME (Ribeiro et al., 2016) often produce inconsistent feature importance rankings. This inconsistency stems from the trade-off between explanatory accuracy and human interpretability (Gilpin et al., 2018). In this work, we use SHAP to explain classification results of our models.

3 Methodology

In this report, we propose a method for predicting emotions from textual data by training a model on a dataset with categorical emotion labels. Further, we train a model on a dataset containing emotion intensity annotations, which enables both the prediction of emotion categories and their intensities. This approach formulates the prediction as a multi-label classification problem within a supervised learning framework. In addition to evaluating and testing the model on our dataset, we compare its performance with human annotations on the same samples. To provide deeper insights into the decision-making process of our models, we use SHAP to analyze interpretability and explainability of our model.

3.1 Datasets

We use the SemEval 2025 Task 11 Track A training and testing dataset (Codabench, 2025) for the English language to train and evaluate our models, respectively. The train dataset contains 2768 samples, the test dataset 116 samples. Each sample consists of an ID, a text, and a set of emotion labels. This set can contain the emotions *anger*, *fear*, *joy*, *sadness* and *surprise* or none of them. Since the dataset contains text samples without any emotion assignment, we decide to add the category *none* and treat it as a class for the absence of any emotion, which leads to a total of six emotion classes.

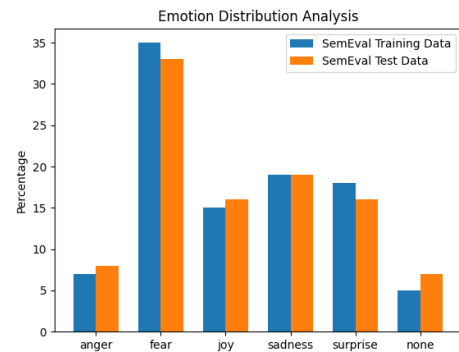


Figure 2 The distribution of emotions in the SemEval dataset for Track A. *Fear* dominates in train and test, followed by *sadness* and *surprise*.

As displayed in Figure 2, the distribution of emotions in the SemEval datasets is unequal. An analysis of the number of distinct labels per sample is visualized in Figure 3.

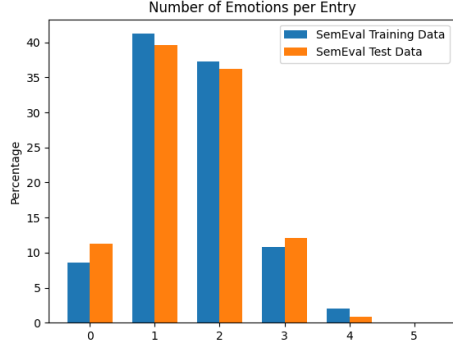


Figure 3 The number of emotions for each sentence in the SemEval dataset for Track A. While most of the samples contain one or two emotions, only a small number has more than three.

To classify both the emotions and their intensities, we utilize the SemEval 2025 Task 11 Track B dataset for English language. This dataset contains the same five emotions as well as three intensity levels, including "none", resulting in 16 different possible classifications. The train dataset contains 2768 samples, the test dataset 116 samples.

3.2 Preprocessing & Tokenization

For emotion classification, we decide to convert the emotion labels to a multi-hot encoded vector of length six, which is the number of distinct emotion classes. For intensity classification, we decide to convert the emotion labels and their intensities to a multi-hot encoded vector of length 16, where we define a single class for each combination of a given emotion and their intensity. Since the emotion *none* does not have an intensity at all, and we get three distinct intensity classes for each of the five emotions, we obtain a total of 16 distinct classes. We preprocess the data using regex and basic string processing methods to remove special characters, URLs and mentions of other users, i.e. strings starting with @. For both tasks, the textual data is tokenized using the pretrained *RobertaTokenizer* with a maximum number of 512 tokens and activated padding. Further, we allow truncation in case the textual input exceeds the length limit.

3.3 Models

A critical part of learning a capable emotion prediction model is the selection of the right model architecture. To be able to test different architectures and their capabilities, we propose five

different models. All models use RoBERTa (Liu et al., 2019) as a backbone and are distinguished by their classification heads. We propose three models that apply pooling to all contextualized token embeddings of RoBERTa’s last hidden layer before they are classified by using fully-connected layer. We will refer to these models as *mean*, *max* and *attention*, named after the applied pooling strategy. For a detailed model description, we refer to Figure 4. In addition, we propose two models that only use the contextual embedding from the CLS token of RoBERTa’s last hidden layer. Both of these models classify this embedding by using fully-connected layers. One model, to which we refer as *deep*, has a deeper classification layer compared to the other one, to which we refer as *pure*. A detailed description of these models can be found in Figure 5. We train and evaluate these models for both Task A and Task B. To meet the requirements of more classification categories in task B, we adjust the dimensionality of the output layer. We compare our models for Task A to a given baseline model (de Kock, 2025), which is a simple bag-of-words model trained on the same SemEval dataset.

For all models, we initialize the weights of RoBERTa with the [pretrained weights from Hugging Face](#). The weights of the classification heads are initialized using the Kaiming-Uniform method (He et al., 2015). All models are trained using 5-fold-cross-validation with a batch size of 16 and a learning rate of $1e - 5$. For the first task we train the models for 8 epochs, for the second task for 16 epochs. We further use the AdamW optimizer (Loshchilov, 2017) with a weight decay of 0.01 and use a weighted random-sampler to mitigate class imbalance during training. Since we are dealing with a multi-label classification problem, we decide for binary-cross-entropy loss with logits (BCEWithLogitsLoss) as a loss function due to numerical stability.

3.4 Evaluation Metrics

Since we are dealing with a multi-label classification problem, we use the *micro*- and *macro-F1* score as the main performance measurement metrics. Further, we provide the *micro*- and *macro*-scores for *precision* and *recall*, since they are also given by the baseline model. In micro-evaluation, the predictions for all labels are aggregated before the metrics are calculated, focusing on over-

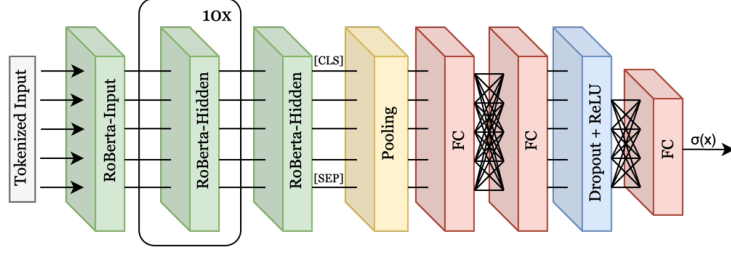


Figure 4 Architecture of the models that use a pooling approach. The tokenized input is passed through the 12 hidden layer of RoBERTa-base. A pooling layer, which can be either *mean*, *max* or *attention* pooling, is then applied to all contextualized token embeddings of the last hidden layer and fed into two fully-connected layers. Before the prediction is made in the output layer, we apply a dropout layer (Srivastava et al., 2014) with $p = 0.1$ and ReLU (Nair and Hinton, 2010) activation.

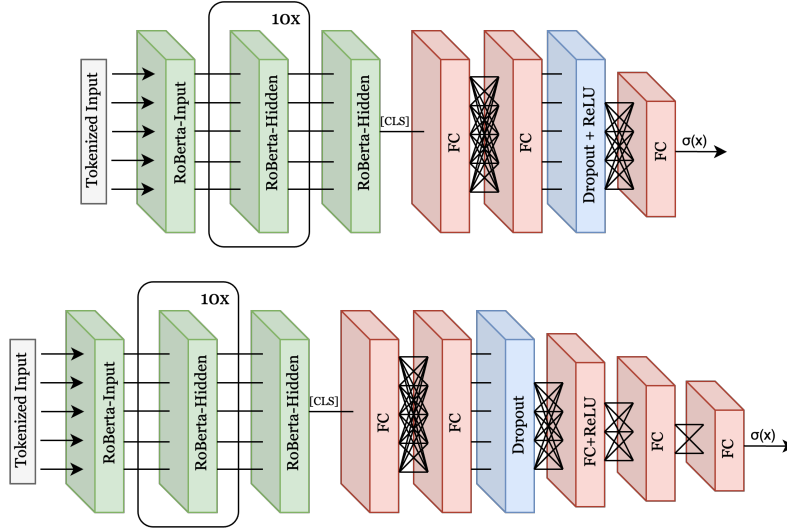


Figure 5 Architecture of the models that only use RoBERTa’s CLS token for classification. While the *pure* model (top) has only two fully connected layers with a dropout layer ($p = 0.1$), the *deep* model (bottom) features four fully connected layers with a dropout layer ($p = 0.1$) after the first one. Both models apply ReLU activation once during the classification procedure.

all performance, whereas in macro-evaluation, the metrics are calculated per label and then averaged, treating all labels equally.

3.5 Human-Machine Evaluation

For comparing the emotion detection capabilities of the models to human performance, we make use of a majority-voting ensemble (Hansen and Salamon, 1990). Therefore, we use inference for every model and then take the majority answers from our models to determine a final classification result and ask a human to assign emotions to a given sample. Both the human and the models are allowed to assign multiple of the six emotion

categories explained in Section 3.1. Both answers are compared to the ground truth. We decide to use examples from the SemEval Task A test dataset for this human-machine competition. The performance is then evaluated by calculating the *macro*-, *micro*- and *weighted-F1* score. In our experiment, we use 15 randomly-selected text phrases from the test set.

3.6 Explainability

We compute and visualize SHAP values using the Python SHAP library¹. The explainer perturbs inputs using the entire training dataset. To ensure sufficient token diversity, the train dataset is used,

¹github.com/shap/shap

as the smaller test dataset alone would not suffice. Due to the high computational cost of SHAP and the diminishing returns of additional model comparisons, we analyze only the *max* and *mean* models, as they demonstrate strong performance in the emotion detection task.

4 Evaluation

The following section will outline the results for our proposed methods for both emotion classification in Section 4.1 and emotion intensity classification in Section 4.2. Further, we will discuss the experimental outcomes of the human-machine competition in Section 4.3 and explainability of the models in Section 4.4.

4.1 Emotion Classification

For an analysis of our models, we use the best performing model of each category across all performed folds.

	Baseline	Attention	Deep	Max	Mean	Pure
Precision	39.81	74.74	74.88	68.50	76.86	78.54
Recall	64.61	75.13	70.71	79.44	76.93	70.67
F1-score	47.85	73.29	71.41	73.15	75.74	73.45

Table 1 The performance of the best performing models across all folds in percent (%) using *macro-evaluation*. While all models outperformed the *baseline*, the *mean* model has the best overall f1-score with 75.74%. The best precision is achieved by the *pure* model (78.54%). The highest recall is reached by the *max* model (79.44%).

All our proposed models outperform the *baseline* model up to 27.89% in macro-, and 20.94% in micro-evaluation. Further, we see that our best performing model (*mean*) has a 4.33% better F1-performance in macro-evaluation compared to our worst performing model (*deep*). This gap is reduced towards 2.52% with micro-evaluation. A detailed analysis of the achieved performance is depicted in Table 1 for macro-, and Table 2 for micro-evaluation. The results indicate that the *max* model is superior if one does not want to miss out the recognition of any emotion, since it selects the most informative token representations. The *pure* model in contrast is superior for precise emotion predictions, since it only relies on the CLS token which captures the global information of the input text. The best overall model which balances both

strengths is the *mean* model, which averages the weighting of all retrieved tokens.

	Baseline	Attention	Deep	Max	Mean	Pure
Precision	44.41	70.67	71.78	70.45	73.07	74.47
Recall	74.47	77.78	76.72	82.01	80.42	74.07
F1-score	55.63	74.05	74.16	75.79	76.57	74.27

Table 2 The performance of the best performing models across all folds in percent (%) using *micro-evaluation*. While all models outperformed the *baseline* in terms of f1-score, the *mean* model has the best overall f1-score with 76.57%. The best precision is achieved by the *pure* model (74.47%). The highest recall is reached by the *max* model (82.01%).

4.2 Intensity Classification

To analyze the performance of our models, we use the best performing model of each category across all performed folds. Further, we restrict ourselves to the micro-evaluation method. It can be seen that the models lack far behind their emotion detection capabilities when they are faced with the additional task of intensity recognition, as displayed in Table 2.

	Attention	Deep	Max	Mean	Pure
Precision	52.23	51.70	45.59	45.96	50.34
Recall	43.39	40.21	49.21	48.15	39.68
F1-score	47.40	45.24	47.33	47.03	44.38

Table 3 The performance of the best performing intensity models across all folds in percent (%) using *micro-evaluation*. The *attention* model has the best overall f1-score with 47.40%. The best precision is also achieved by the *attention* model (52.23%). The highest recall is reached by the *max* model (49.21%).

The *attention* model performs best in F1-score. This might indicate that the learned weighting of the contextualized tokens from the last layer benefits in predicting more nuanced emotion categories like different intensities. While the *mean* and *max* models still excel in their recall scores, they lack in precision, which hints that averaging the token embeddings is not the most efficient operation if we want to distinguish precisely between nuanced categories.

4.3 Human-Machine Evaluation

We use our best models of each category to evaluate them against human performance. The results of this experiment are displayed in Figure 6. The

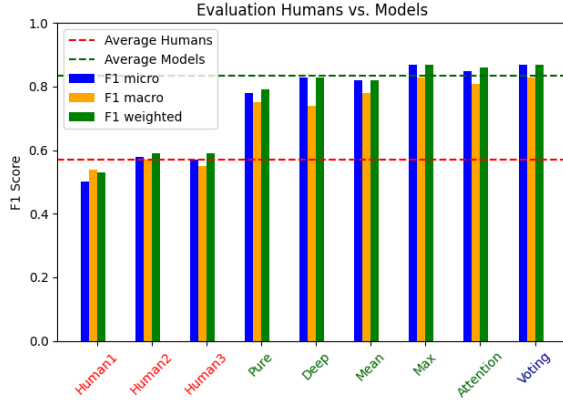


Figure 6 The performance of three humans and our models when classifying 15 random sentences from the SemEval test dataset. The averages are calculated using the weighted F1-Score. The average weighted F1-Score for the humans is 0.57, while it is 0.83 for the models.

performance of our five models is significantly better (26% on average) than the performance of the humans. We draw two conclusions from this small experiment. Firstly, it seems that our models were able to learn how to detect emotions from text, since they are able to correctly classify most sentences in the dataset. Secondly, humans, who are typically expected to accurately detect emotions, performed worse than the models. This may be because the SemEval dataset contains minimal or no contextual information for the text samples, making it more challenging for humans to interpret emotions accurately. A more complex evaluation would be needed to determine the validity of our results and our conclusions.

4.4 Explainability

We use both global and local explainability methods to analyze the classification decisions of our models. Global methods identify the tokens that most strongly influence classification across the dataset, potentially revealing biases unrelated to the data or context. An example of global analysis can be seen in Figure 7 for the *max* and in Figure 8 for the *mean* model. Among the top 20 tokens contributing to emotion

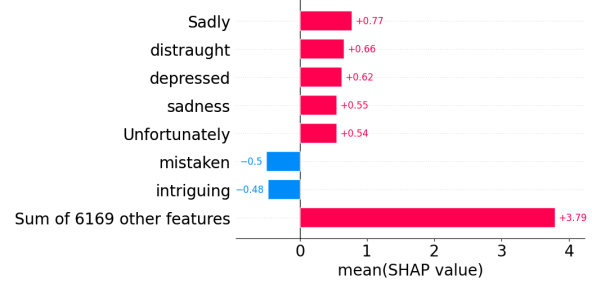


Figure 7 Most important words for *sadness* classification of the *max* model identified by SHAP.

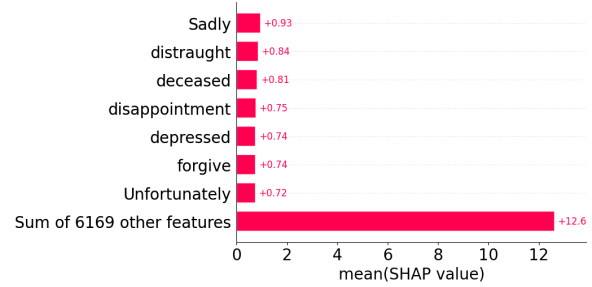


Figure 8 Most important words for *sadness* classification of the *mean* model identified by SHAP.

classification for each of the emotions, about 7% of the tokens appear to be implausible by human judgment, indicating possible inconsistencies in the model's decision process. For example, the token "*Charlie*" received the 4th (*max*) or 10th (*mean*) highest SHAP score for *surprise*, even though there is no obvious semantic connection. Further, the confidence in the predictions of our models can also be increased by comparing the key features identified by the *mean* and *max* model. If both models emphasize similar and interpretable features, this indicates convergence on meaningful patterns. However, our global explanations reveal considerable discrepancies between the models, which reduces the reliability of this approach for their explainability.

Local explanations assess the significance of each token within a specific sentence, providing insight into how the model interprets words in context. Force plots illustrate these interpretations, as shown in Figure 9, where the *mean* model associates "*respected*" with *sadness* when paired with *distraught*, despite "*respected*" typically carrying a positive meaning. This local analysis is particularly valuable for understanding mis-

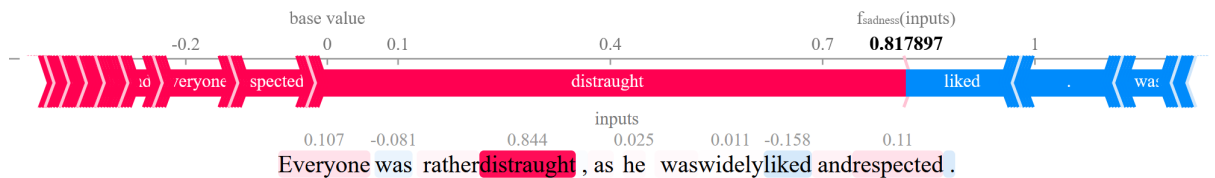


Figure 9 A local force plot illustrating the explanation for *sadness* in a single sentence. The *mean* model predicts sadness with 82% confidence, where red tokens increase and blue tokens decrease the classification probability. Token contributions are highly context-dependent; for example, "respected" may convey a positive meaning in a different sentence.

classifications. However, when the model fails to recognize an emotion, most force plots remain empty due to a lack of contributing tokens.

In summary, we observe that SHAP values can be valuable for providing localized explanations for classification results. However, due to their example-based nature, their usefulness for establishing overall confidence in our models is limited.

5 Discussion

Our evaluation finds that even humans encounter difficulties when classifying short sentences with minimal context, highlighting the complexity of emotion classification. Our analysis of the dataset reveals an imbalanced distribution of labels. Further, some label assignments in the data do not seem intuitively justifiable. This raises concerns about ground-truth reliability, highlighting the need for a more balanced dataset and an explainable labeling strategy to improve training success.

A major limitation is the size of the dataset, which may not be sufficient to effectively train large models. This can be seen by the deep model, which performed worse compared to smaller models, suggesting that a larger dataset could improve weighted learning in deeper architectures. Furthermore, incorporating emotion lexicons and utilizing larger transformer models such as RoBERTa-large could improve both emotion and intensity prediction. Adapting the architecture to include distinct heads for emotion and intensity classification could further refine the predictions, especially for emotion intensity.

The reliability of the human-machine evaluation could be limited by the small number of test

subjects. Increasing the number of participants and using more diverse test data would allow a more meaningful comparison between model and human performance. While our approach provides initial insights into performance differences, further studies with larger samples are needed for more stable results.

The inconsistency of SHAP-based explanations across models presents a challenge in model interpretability, limiting their ability to enhance trust and confidence in model reliability, which is crucial if such models should be included in medical applications like digital-health monitoring.

Further, the use of text-to-emotion models raises ethical concerns about privacy, bias and misinterpretation. As we already indicated previously, users need to be aware that these models may reflect biases in their training data, which can lead to inaccurate or unfair emotional evaluations. Thus, it is important to ensure transparency as well as a consent and responsible use to avoid harm or manipulation and strengthen the user's confidence in the model.

6 Contribution statement

- **Elia Rühle** - Architecture Design, Model Training, Related Work Research, Writing
- **Charles Krüger** - Evaluation, Dataset-Analysis, Model Training, Writing
- **Simon Crämer** - Explainability, Writing
- **Michael Färber** - Writing, Supervision
- **Simon Rasniewski** - Writing, Supervision

The repository for our project code: <https://github.com/kruegercharles/LLM-SemEval>

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